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Final Year Design Project Proposal

Project Title:

Design and Development of a Communicable Fault Passage Indicator for Low-Voltage Distribution Networks in Bangladesh.

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Design and Development of a Communicable Fault Passage Indicator Device for Low Voltage Distribution Networks in Bangladesh.

CHAPTER 1: DEFINING THE CHALLENGE: PROBLEM, SCOPE, AND OBJECTIVES

1.1) Problem Statement and Background Research (CO1)

Problem Statement:

Bangladesh's low-voltage (LV) distribution network which serves mainly residential and commercial consumers face frequent faults. Bangladesh's distribution network experiences technical (10–15%) and non-technical (5–10%) losses (According to DPDC) despite sufficient generation capacity, indicating inefficiencies at the distribution level. The LV networks remain largely unmonitored, relying on manual inspections and line patrols for fault localization. This leads to long restoration times of about 3-4 hours on average [1]. These outages are caused due to old conductors, overloading, illegal connections, high-impedance faults and environmental disturbances [2]. Additionally, the lack of accurate load forecasting and real-time monitoring makes it challenging for utilities to predict faults and optimize grid management. Prolonged fault detection contributes to high SAIDI (System Average Interruption Duration Index) values (12–15 hours annually), particularly in rural distribution networks. (DPDC Annual Report 2023-2024). It significantly increases ENS (Energy Not Supplied) and the associated economic cost in $/kWh of unserved energy, often dominating the annual reliability penalties imposed on utilities [3], [4].

Background Research:

From a hardware standpoint, many MV (medium voltage) and LV feeders still rely on sparse protection relays and a limited number of field devices, so a single missed or false indication can expand the isolation area, raising the number of affected customers and the total ENS [3], [4], [8]. Reviews of distributed-generation‑integrated networks emphasize that hardware‑based protection schemes must operate correctly under wide variations in short‑circuit level; otherwise, protection blinding and unnecessary trips increase both direct interruption costs and regulatory penalties [9]. As utilities seek to minimize capital and operational expenditure, there is strong motivation to deploy low‑power, environmentally robust fault‑sensing hardware with high mean‑time‑between‑failures (MTBF), modest per‑unit cost and dependable fault indication to support fast, selective restoration [3], [4], [8], [9].

Conventional LV protection relies heavily on iron‑cored current transformers (CTs) feeding overcurrent and earth‑fault relays. While mature and widely standardized, CTs are bulky, require primary circuit interruption for installation and can saturate under high fault currents or DC offset. This causes measurement distortion and compromised fault detection and location [3], 2, [8]. CT errors of several percent in amplitude and phase under transient or high-burden conditions are common, limiting the achievable accuracy of impedance-based or directional elements that depend on precise current phasors [3], [4].

Fault passage indicators (FPIs) are dedicated field devices that monitor local line conditions to indicate the passage of fault currents. Modern FPIs incorporate various sensing hardware: clamp-on CTs or Rogowski coils for current, capacitive or electric-field probes for voltage and integrated microcontrollers with low-power communication modules [3], [8], [13]. In resistor-grounded MV networks, earth faults are frequent but can have relatively low zero-sequence currents, challenging classical FPI pickup criteria. An adaptive zero-sequence overcurrent criterion for FPIs demonstrated that, by adjusting pickup thresholds and time settings based on network conditions, detection of earth faults, especially high-impedance faults, can be improved by more than 40% compared with classical fixed-threshold criteria [16]. This highlights that even with conventional sensors, intelligent signal processing in FPI hardware can significantly improve detection sensitivity.

Alternative non-contact FPI-like concepts based on capacitive probes mounted near conductors have been proposed to detect zero-sequence voltage components and asymmetry without direct electrical connection [13]. Laboratory prototypes have shown that suitably arranged capacitive probes can detect earth faults with sensitivity comparable to conventional VT-based systems [13]. However, moving from prototypes to outdoor hardware requires addressing issues like stable calibration under temperature and humidity, robust mechanical design, adequate IP rating for rain and dust and ensuring sufficient immunity to overvoltages and electromagnetic interference [3], [13], [16].

Hall-effect current sensors (HECSs) and related integrated magnetic sensors offer galvanic isolation, wide bandwidth, and compact form factors, with typical linearity on the order of ±0.5–1% and dynamic ranges from a few amperes to kiloampere levels [10]. Implementation studies highlight their advantages in terms of low power consumption (tens of mW for integrated CMOS Hall front-ends) and suitability for embedded or retrofit applications in power electronics and smart-grid environments [10]. However, Hall sensors are sensitive to temperature and external fields; temperature drift and offset errors can degrade accuracy, especially when operating over industrial ranges of −40 °C to +85 °C [10]. Shielding and calibration circuits increase complexity and cost, and long-term stability and MTBF under outdoor MV conditions (humidity, vibration, pollution) remain less documented in field studies compared with traditional CTs [3], [4], [10]. In resource-constrained utilities, the need to balance per-unit cost, installation complexity, insulation coordination, and accuracy has slowed the systematic replacement of CT-based hardware, particularly on overhead rural feeders [3], [8].

Rogowski coils demonstrated high signal-to-noise ratio and reliable measurement of high-frequency components, outperforming high-frequency current transformers for partial discharge (PD) detection in MV/HV air-insulated switchgear [11]. Being lightweight and flexible, Rogowski coils can be retrofitted around existing conductors without interrupting service and their inherently low output voltage contributes to intrinsic safety. However, Rogowski coils measure only the derivative of current and require integrating electronics, which must maintain linearity and low noise over large dynamic ranges, including from low load currents up to tens of kA of fault current [11]. Sensitivity to mutual coupling and external electromagnetic interference complicates installation in compact MV switchgear and cable terminations [11], [12]. Non‑contact magnetic sensors, such as external loop antennas or D‑dot/E‑field probes, have been explored for PD and transient detection. These sensors offer fully non‑intrusive operation and good high‑frequency response but typically provide only relative or uncalibrated measurements without careful calibration, limiting their use for quantitative fault-current estimation or impedance‑based location [11], [12]. For field deployment on overhead lines and outdoor equipment, environmental protection (e.g., IP67 ingress rating) and mechanical robustness under wind and contamination are essential, yet many research prototypes are characterized only in laboratory or indoor test bays [11]–[13].

Fault‑sensing hardware research is geographically diverse. In Europe, work on capacitive probes for LV earth‑fault detection and on improved FPI criteria for resistor‑grounded networks has focused on overhead lines typical of Central and Eastern European grids [13], [16]. Hybrid PD sensing for LV switchgear has been driven by European utilities and manufacturers concerned with asset health and life extension [11].

In developing countries and emerging smart cities, research highlights the mismatch between sophisticated sensor/communication solutions and the financial capabilities of local utilities; low‑cost, low‑power, easily deployable devices are repeatedly identified as a priority, especially for rural MV feeders with few existing measurement points [3], [4], [5], [9], [18].

Research Gap:

Across hardware categories, several recurring limitations constrain accurate fault detection and location in MV networks. Conventional CT/VT‑based solutions require outages and significant installation work [3], [8], [12]. This limits widespread deployment on existing networks. High‑precision sensors offer better linearity and bandwidth but at per‑unit costs and power budgets that are difficult to justify for dense field deployment [8], [10], [11], [17]. Many experimental capacitive and non‑contact sensors are validated only in laboratories. Long‑term MTBF, drift and failure modes under outdoor MV conditions (pollution, UV, vibration, −40 °C to +85 °C) are seldom reported [11]–[13], [16], [18]. Next, rural and suburban feeders often have very few measurement points. Sophisticated multi‑sensor or PMU‑based platforms assume an instrumentation density and communication infrastructure that do not exist in these grids [3], [4], [5], [18]. Pole‑top or line‑mounted devices must operate on constrained power budgets, sometimes from batteries or line energy harvesting while supporting reliable communication (e.g., PLC, sub‑GHz RF, LTE‑M) over long distances. Many high‑performance sensing concepts have not been engineered under such constraints [3], [5], [12], [18].

These gaps collectively point to the need for compact, low‑power, environmentally hardened sensing hardware capable of providing dependable fault passage and direction information without the complexity and cost of full PMU‑class instrumentation or dense multi‑sensor arrays.

Furthermore, electricity theft, including practices like meter tampering, meter bypassing, and illegal line hooking remains a significant challenge that manual inspection systems are ill-equipped to detect. In Bangladesh, these illegal activities not only result in financial losses but also contribute to the unavailability of power for legitimate consumers due to overloading caused by unauthorized connections. The Bangladesh Energy Regulatory Commission (BERC) reports that around 15% of electricity consumption is unaccounted for due to theft, costing the economy around $600 million annually.

Rationale:

Given the central role of accurate fault indication in minimizing ENS and reliability penalties, and the persistent constraints on CAPEX and OPEX for many MV operators, there is a strong rationale for focused research on hardware‑efficient fault sensing specifically tailored to distribution environments. Existing work provides high‑end solutions (PMUs, multi‑sensor switchgear monitors) and innovative but immature concepts (capacitive probes, electric‑field sensors) leaves a substantial space between simple mechanical FPIs and complex, communication‑intensive platforms [3], [8], [11]–[13], [16], [18].

For a final‑year design project, concentrating on the hardware layer such as sensing elements, analog front‑ends, power management and environmental design offers a realistic yet impactful scope. It allows exploration of trade‑offs among sensor accuracy, linearity, bandwidth, power consumption (targeting tens to a few hundred mW), communication options (PLC vs. low‑power RF), and expected MTBF, informed by documented limitations in current devices and deployment practices [3], [4], [8], [10]–[13], [16], [18].

Within the broader hardware landscape, fault passage indicators occupy a favorable point in the cost–functionality spectrum. Compared with substation‑centric CT/VT/relay schemes and MV‑level PMUs, FPIs can be non‑intrusive. They can clamp‑on CTs or Rogowski coils and capacitive probes can be installed on live lines or cables without primary disconnection, reducing installation cost and service disruption [8], [11], [13], [16]. Modern FPI designs can operate from harvested line energy or small batteries with power budgets in the tens of mW, far below that of full PMU platforms, simplifying deployment on remote feeders [3], [5], [15], [18]. Compact, sealed enclosures (IP65–IP67) and industrial temperature ratings are easier to achieve with simple analog/digital electronics than with complex multi‑board PMU or communication racks. This supports higher MTBF in harsh outdoor conditions [11]–[13], [16]. Furthermore per‑unit device costs can be kept low enough to justify installation at multiple points along long rural feeders, improving fault‑section resolution without requiring full measurement coverage or high‑bandwidth communications [3], [4], [8], [16], [18].

Given these research findings, it is clear that Bangladesh's LV distribution infrastructure would benefit from an FPI system that incorporates advanced fault detection algorithms, real-time communication and IoT-based monitoring. Such a system would not only enhance fault detection and service restoration but also significantly reduce electricity theft by integrating monitoring capabilities to detect abnormal current behavior indicative of illegal connections. This would address the gap between the existing manual, inefficient fault detection practices and the growing demand for automated, responsive solutions in LV networks, ultimately reducing operational costs and improving system reliability.

1.2) Scope and Objectives (CO1)

Scope:

This project focuses on designing, developing and testing a real-time communicable Fault Passage Indicator (FPI) device for low-voltage distribution networks in Bangladesh. The solution will cover the development of the hardware system, fault detection algorithms and IoT-enabled communication for interfacing the FPI device with a central monitoring server. It will include laboratory testing under simulated low-voltage fault conditions to evaluate the system's accuracy, reliability and performance. The final output will be a working prototype of the FPI device.

The scope excludes field deployment and long-term operational testing, which will be addressed in future work. Instead of pursuing a massive rollout across the DPDC's entire infrastructure, the effort is focused on a functional prototype validated through a limited number of test feeders. We've also chosen to bypass complex automation like FLISR systems and the in-house fabrication of semiconductors. We have opted instead for proven commercial components like to ensure reliability. Safety and accessibility remain a priority, so physical installation on live 11kV lines and extensive personnel training programs have been set aside in favor of thorough documentation and controlled testing. Additionally, the project's lifecycle is strictly defined. It will be concluded after a twelve-month maintenance window.

Chapter for 11KV Distribution Network in Bangladesh:

Bangladesh's 11 kV distribution network, spanning over 62,000 kilometers of predominantly overhead lines managed by utilities such as Dhaka Power Distribution Company Limited (DPDC), Dhaka Electric Supply Company Limited (DESCO), Bangladesh Power Development Board (BPDB) and West Zone Power Distribution Company Limited (WZPDCL). These represent a critical yet challenged infrastructure supporting the nation's rapid electrification. This chapter delineates the precise scope of a proposed FPI installation initiative tailored to Bangladesh's 11 kV radial topology. This situates it within the broader imperatives of grid modernization under the government's Power Sector Master Plan and Asian Development Bank-funded efficiency enhancement programs. By focusing exclusively on fault indication rather than automation or protection, the scope aligns with the operational constraints of resource-limited utilities, enhancing System Average Interruption Duration Index (SAIDI) compliance.

The 11 kV network in Bangladesh exhibits pronounced vulnerability to single-line-to-ground (SLG) faults constituting approximately 70% of incidents. This is exacerbated by monsoon-induced tree contacts, lightning strikes averaging 150 thunderstorm days annually and conductive pollution accumulation on insulators under relative humidities routinely surpassing 85%. Urban feeders in DPDC and DESCO territories, serving densely populated load centers like Dhaka with peak demands exceeding 1,200 MW per feeder, contend with high impedance earth faults (HIEFs) below relay pickup thresholds (typically 150-200 A). On the other hand, rural WZPDCL and BPDB lines often spanning 20-30 km with minimal lateral sectionalizing face prolonged outages due to sparse crew availability and navigational challenges amid flood-prone terrains.

Conventional fault management relies on upstream recloser or fuse operation followed by line crew patrolling, a labor-intensive process yielding SAIDI contributions of 4-6 hours per major event, compounded by the absence of real-time SCADA granularity at the distribution level. This paradigm persists despite national smart grid pilots, as full feeder automation remains cost-prohibitive given average FPI unit economics of BDT 20,000-40,000 versus imported alternatives at BDT 1,00,000+.

The scope can encompass strategic deployment of 500-1,000 FPIs across 50 high-outage 11 kV feeders, prioritizing DPDC's urban F-1 to F-20 clusters and WZPDCL's rural Khulna spans exhibiting SAIDI exceedances greater than 20%. Installation targets lateral branch points and high-fault sections identified via historical outage logs, with each FPI configured for bidirectional fault discrimination on tapped feeders, sensing transients from 40-500 A over 50-100 ms durations inclusive of HIEFs and conductor clashing. Integration entails GSM/4G telemetry to existing OMS platforms, furnishing control rooms with geolocated fault vectors reducible to 500-meter patrol radii, alongside local LED/buzzer actuation for line crew validation.

Fault data aggregation supports dashboard visualization of MTTD reductions from 180 minutes to under 30 minutes, with accuracy benchmarks of 90% laboratory and 85% field detection across six fault archetypes validated under IEEE C37.113-equivalent protocols. Conversely, the initiative expressly excludes protection coordination, eschewing recloser interfacing or zone-selective interlocking to preserve existing electromechanical relay schemes. Automatic fault isolation, remote switching, or voltage-based autoreclosing hallmarks of full FLISR (Fault Location, Isolation, Service Restoration) fall beyond purview, as do primary overcurrent/earth-fault relaying functions that might necessitate utility-wide settings revisions. Pole reinforcement or conductor replacements remain external, with FPIs designed as retrofit clamps accommodating 50-300 mm² ACSR Panther/Zebra conductors ubiquitous in Bangladesh.

Applicability spans urban feeders in DPDC/DESCO (e.g., Mirpur-Motijheel corridors with 4-6 laterals per 15 km), semi-urban BPDB clusters in Chattogram, and rural WZPDCL lines exceeding 25 km sans intermediate RMUs. In Dhaka's conurbation, FPIs mitigate urban heat island effects amplifying conductor sag (up to 7 meters under 45°C peaks) and pollution flashovers, while rural deployments counter monsoon flooding submerging pole bases and cyclone-induced tree falls. Power autonomy via CR2477 lithium cells (10-year shelf life) or 5W solar obviates tapping constraints on unpaved access roads, with maintenance scoped to annual visual inspections aligning with existing vegetation management.

This demarcation preserves operational workflows. Dispatchers receive FPI alerts superseding vague relay annunciations, crews prosecute targeted patrols versus blind sectionalizing, and post-mortems leverage logged transients for root-cause analytics without procedural overhauls. Reliability indices accrue via compounded MTTD compression, projecting 60-70% SAIDI abatement on instrumented feeders per IEC 61400-24 lightning exposure models adapted to Bangladesh's 120 kV.

Embedded within Bangladesh's Smart Distribution Roadmap and REB's digital transformation under PSDP-2041, FPI deployment constitutes a pragmatic precursor to advanced metering and substation automation, augmenting situational awareness sans the fiscal barrier of comprehensive SCADA/ADMS. By focusing on indication, the scope catalyzes cultural shifts toward data-driven outage stewardship, positioning utilities for RDSS (Revamped Distribution Sector Scheme) incentives while mitigating the 2-3% GDP encumbrance of reliability deficits.

The circumscribed scope of FPI installation on Bangladesh's 11 kV network proffers a surgically precise remedy to fault localization latencies, harnessing proven sensing and telemetry paradigms to indigenous exigencies without encumbering extant protection architectures. This delineation ensures feasibility, scalability, and measurability, furnishing a foundational augmentation to the nation's distribution resilience.

Objectives:

- Design and development of an intelligent Fault Passage Indicator (FPI) device capable of detecting and identifying fault currents.
- Develop communication capabilities to interface the FPI device with a central monitoring server.
- Verify system performance by conducting laboratory tests across multiple fault scenarios to assess reliability and accuracy against standard benchmarks.

CHAPTER 2: FOUNDATIONS OF THE SOLUTION: DEFINING REQUIREMENTS, SPECIFICATIONS AND CONSTRAINTS

2.1) Requirements, Specifications, and Constraints (CO2)

To achieve its intended purpose, which is known as requirements, the project has specific goals or conditions to be met. For this project, the requirements are mainly divided into two types:

- Functional Requirements: Requirements which refers to what the project is going to do. It describes core attributes needed to fulfill its purpose.
- Non-Functional Requirements: This is about the aspects of how well the project will perform. It consists of factors which contribute to the optimization of the project overall.

a) Functional Requirements

- Detect Fault Currents: Detect abnormal current patterns such as fault currents, high/low impedance faults, and sudden fluctuations.Reduces fault detection identification time.
- Real-Time Monitoring: Perform continuous current monitoring using non-intrusive Hall-effect sensors.
- Distinguish Fault-induced from Normal Load Variation: Identify fault-induced current signatures separately from normal load variations.
- Communicate Fault Alerts: Send fault alerts to a central server via LoRaWAN, GSM, or Wi-Fi communication protocols.
- Provide Local Indication: Indicate faults locally with LEDs or a buzzer during fault events. Sends real time status alerts.
- Current Theft: Detects Current theft by comparing the flow of current in between 2 consecutive poles.
- Support Reset Options: Allow both automatic and manual reset after fault events.

b) Non-Functional Requirements

- Manual Installation: The device should be easy to install manually with minimal tools.
- Fault type : The device detects the type of faults.
- Repeated Faults: Demonstrate detection performance across repeated fault events under identical operating conditions.
- Response time: It should be less than 0.2s to 0.5s.
- Low Power Consumption: Operate efficiently with a rechargeable battery and optional solar charging.
- Compact and Weather-Resistant: Must be compact, weather-resistant (IP55+), and easy to mount on poles or distribution boxes.
- Communication Range: Maintain a communication range of 1–2 km (LoRaWAN) or continuous connectivity via LTE.

- Low Latency: Ensure less than 150 ms latency from fault detection to server update.
- Reliability and Accuracy: Provide reliable and accurate fault detection under various conditions.
- External Parameters:Analyses the wire condition and consider weather conditions.

c) Specifications

- The system operates at a voltage level of **230V/440V**, designed specifically for low-voltage distribution lines.
- The device monitors currents within a **range of 0–100 A**, ensuring comprehensive fault detection across typical load conditions.
- The fault detection system uses Hall-effect sensors with a current measurement range of **±100 A**, offering high accuracy with a resolution of **0.1 A** and an accuracy tolerance of **±1%**.
- The microcontroller, single-board computer (SBC), or FPGA used in the system is capable of handling the required processing power and communication tasks with an operating clock speed of **≥ 100 MHz** and supporting interfaces for communication.
- Communication is facilitated through LoRaWAN, GSM, or Wi-Fi, ensuring reliable data transmission over **distances of 1–2 km** for LoRaWAN or continuous LTE connectivity.
- The power system uses a 3.7V Li-ion or $LiFePO_4$ battery, with an optional **5W–10W solar module** to ensure long-term efficiency and sustainability.
- The system ensures real-time fault alerts with a latency of **less than 150 ms** from event detection to server update.
- Designed to be compact and easy to install, the device can be mounted on poles or distribution boxes, providing ease of use in field installations.

d) Constraints

- Detection Challenges: LV current fluctuations are harder to analyse than MV faults, requiring advanced detection algorithms.
- Fault Time: The fault must sustain for 40 ms to 100ms.
- Communication Reliability: Dependent on local infrastructure like LoRaWAN or LTE availability.
- Safety: Must meet safety standards for operation near exposed LV lines.The high voltage transmission line Electromagnetic interference must be less than 10 Volts per meter (V/m) around the FPI.
- Budget Limitations: Limited budget for advanced industrial-grade sensors and components.

CHAPTER 3: DESIGNING THE SOLUTION: METHODOLOGY AND APPROACHES

3.1) Methodology

This project aims to develop, design and test a smart Fault Passage Indicator (FPI) system that would be able to identify faults in low-voltage (LV) distribution feeders in Bangladesh. The project will specifically be aimed at having a high degree of reliability in fault detection and communication, integrated to a minimal degree to ensure high real-time performance and long lifespan in a variety of environmental settings.

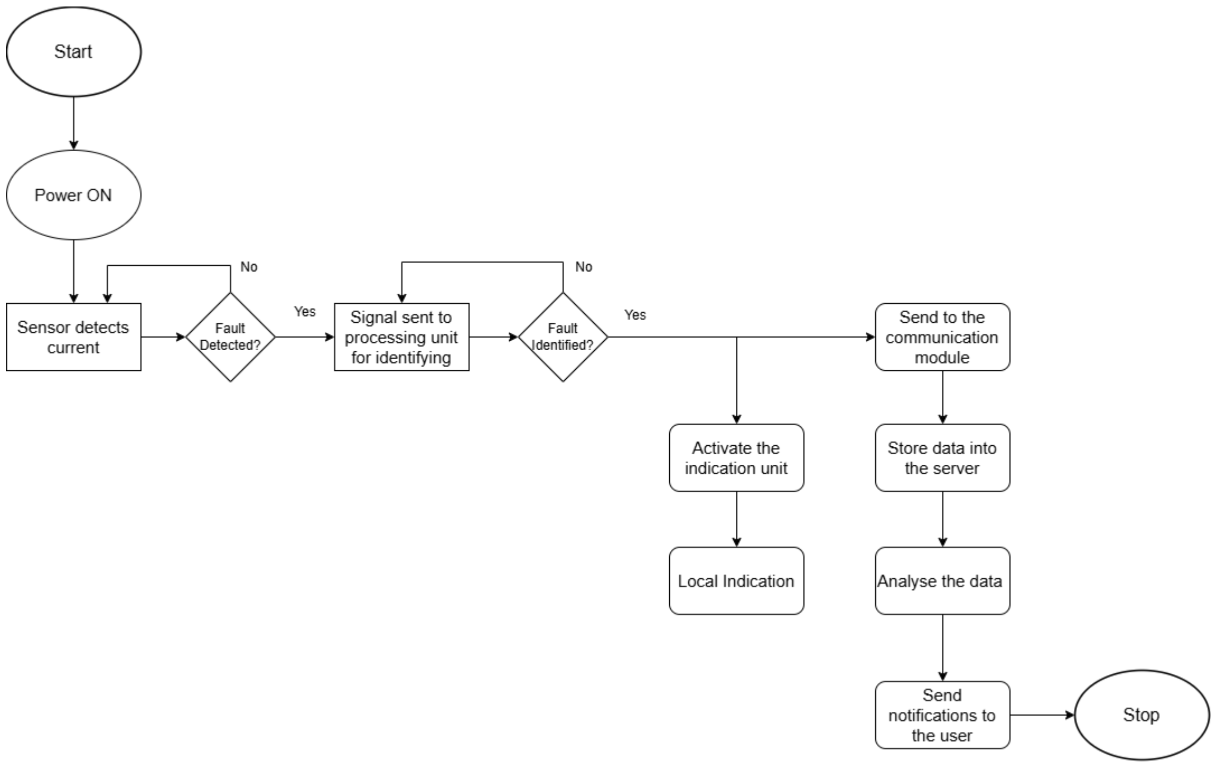


Figure 1: Flowchart of the system

The Power Unit takes part in the system workflow by providing stable and regulated power supplies to all the subsystems to ensure constant and reliable system operation. It also provides energy autonomy in favor of long-term deployment in low-voltage distribution networks.

The Current Sensor Unit continuously monitors the line current to spot the abnormal variations or abnormalities in the current. The information regarding sensed current is transfected to the Processing Unit, where a programmable threshold and time-based algorithm is carried out. The algorithm compares the current with a number of predefined ranges and classifies the operating condition. For currents in the range of 0-80A, the system identifies normal operation. When the current falls between 80-100 A, a warning condition is detected and at more than 100 A current and longer than the response time (0.2-0.5 ms) will confirm a fault condition. This approach guarantees rapid and accurate fault detection with low levels of false triggering (i.e. from transient disturbances).

Once a fault or abnormal condition becomes classified, the Indication Unit is activated to give immediate local indications with visual indicators like LED or auditory indicators like buzzers. These alerts help the field technician to quickly determine the fault occurring and to locate it on site.

Simultaneously, the Communication Unit sends the corresponding fault information such as the current magnitude, and whether the fault is occurring or not, time of occurrence, etc to a central server in real time. This facilitates centralized monitoring and decision-making for better lessening of the fault response and restoration time.

Finally, the Receiving and Display Unit is used to display the transmitted data using a web-based interface. Using statistical methods of analysis, the system processes and displays the fault information in charts, graphs, tables and trend plots to enable operators to monitor fault history, system health and real-time operating conditions. This centralized dashboard allows for informed decision-making and preventative maintenance.

Through seamless integration of all the subsystems, the proposed system helps to ensure rapid fault detection, effective local indication and reliable remote monitoring ultimately for the improvement of reliability, performance and response efficiency of low voltage distribution networks.

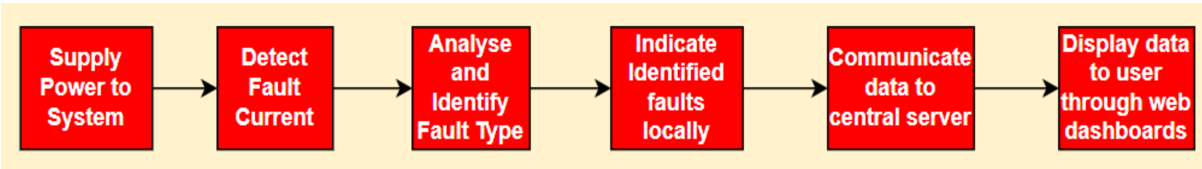
3.2) Multiple Design Approaches

i. Functional Analysis

Basic function: Detect, identify, and communicate faults.

Secondary functions:

- ➢ Supply power to system
- ➢ Detect fault current
- ➢ Analyse and identify fault type
- ➢ Indicate identified fault locally
- ➢ Communicate data to central server
- ➢ Display data to user through web dashboards



ii. Multiple design solutions:

Subfunctions	Approach 1	Approach 2	Approach 3
Supply power to system	CR2032 coin cell	Solar rechargeable battery	
Detect fault current	Rogowski Coil	Hall Current Sensor	Shunt Resistor
Analyse and identify fault type	Microcontroller (e.g, Arduino/STM32 series)	Single Board Computer (e.g, Raspberry Pi)	FPGA
Indicate identified fault locally	LED	Buzzer (Sound/alarm)	
Communicate data to central server	Cellular IoT (e.g, LTE-M, 3G/4G Simcard, GSM Module)	LoRaWAN	Satellite
Display data to the user through web dashboards	Blynk	ThingsBoard	

Table 1: Proposed 3 design approaches for the project

iii. Design Approaches

Design Approach 1:

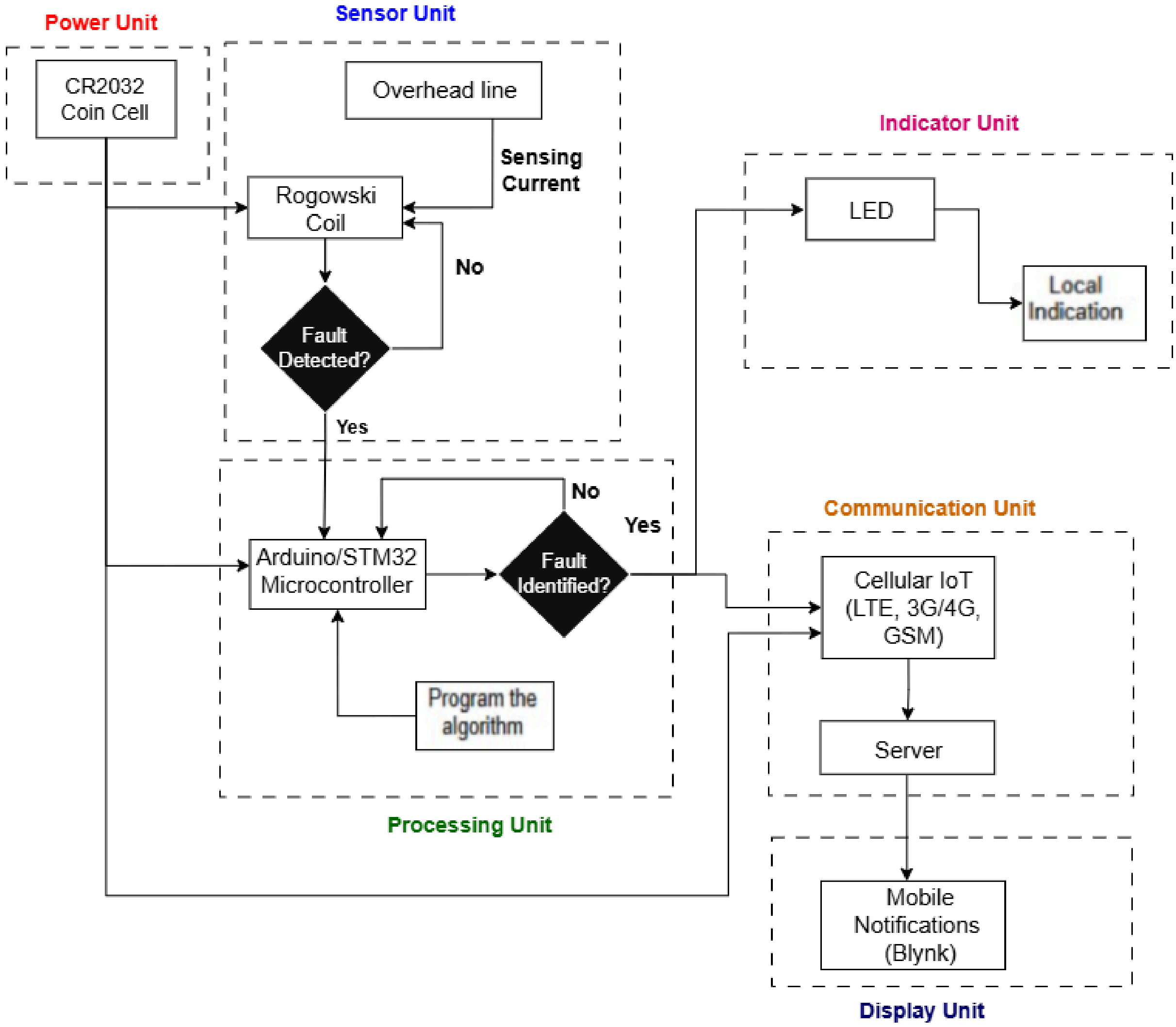


Figure 2: Block diagram for design approach 1

Design Approach 1 follows a simple fault passage indicator which can be used in large-scale implementation with minimum power usage and cost. A non-contact current sensor that is used in this system is a Rogowski coil that generates a voltage in accordance with the change of line current. A signal conditioned via this signal is fed to a microcontroller unit (Arduino/STM32) where integration and threshold based logic is performed to provide an approximation of the current magnitude and fault conditions based on magnitude and both length. Local LED indicators are triggered by a microcontroller when a fault has been detected to give a visual fault indication. At the same time, a GSM module (SIM800L) is used to send fault data including fault status and time of fault to a remote monitoring server. A DC-supported power module is used to supply power and is regulated using a buck converter (LM2596) to provide constant operation. This is a method that engages in the minimization of complexity, practicability, and low price, which is appropriate to be implemented on a large scale in distribution channels.

Design Approach 2:

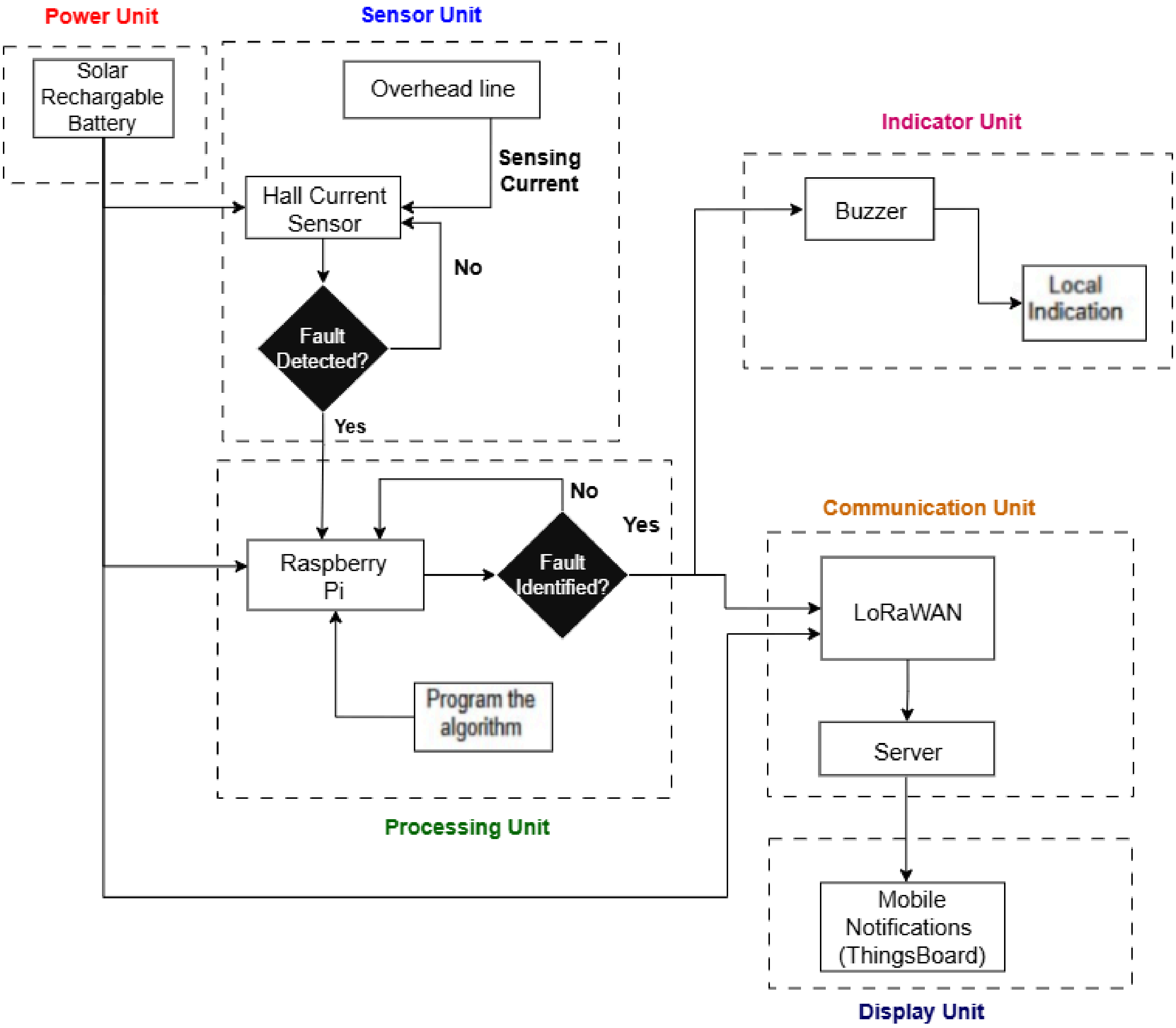


Figure 3: Block diagram for design approach 2

Design Approach 2 is based on the same functional workflow as Design Approach 1, but achieves improved performance and scalability by utilizing higher accuracy sensing and long-range communication. Current measurement is done by using Hall-effect current sensor which gives direct proportional output with respect to line current having more accuracy and immunity against noise. The output of the sensors is then processed with an ESP32 or Raspberry Pi Pico W, allowing for improved signal filtering, fault classification and all edge level data processing. For communication a LoRa transceiver module (RFM95) is used to send the fault data over a long distance with a low power consumption, and hence the system can be deployed remotely or in rural locations. Data may be sent to a gateway to be monitored in a cloud. The power generation is from a solar panel and rechargeable li-ion battery with the charge controller that provides for autonomous use. This design is a mid-scale solution which balances cost, performance and communication range.

Design Approach 3:

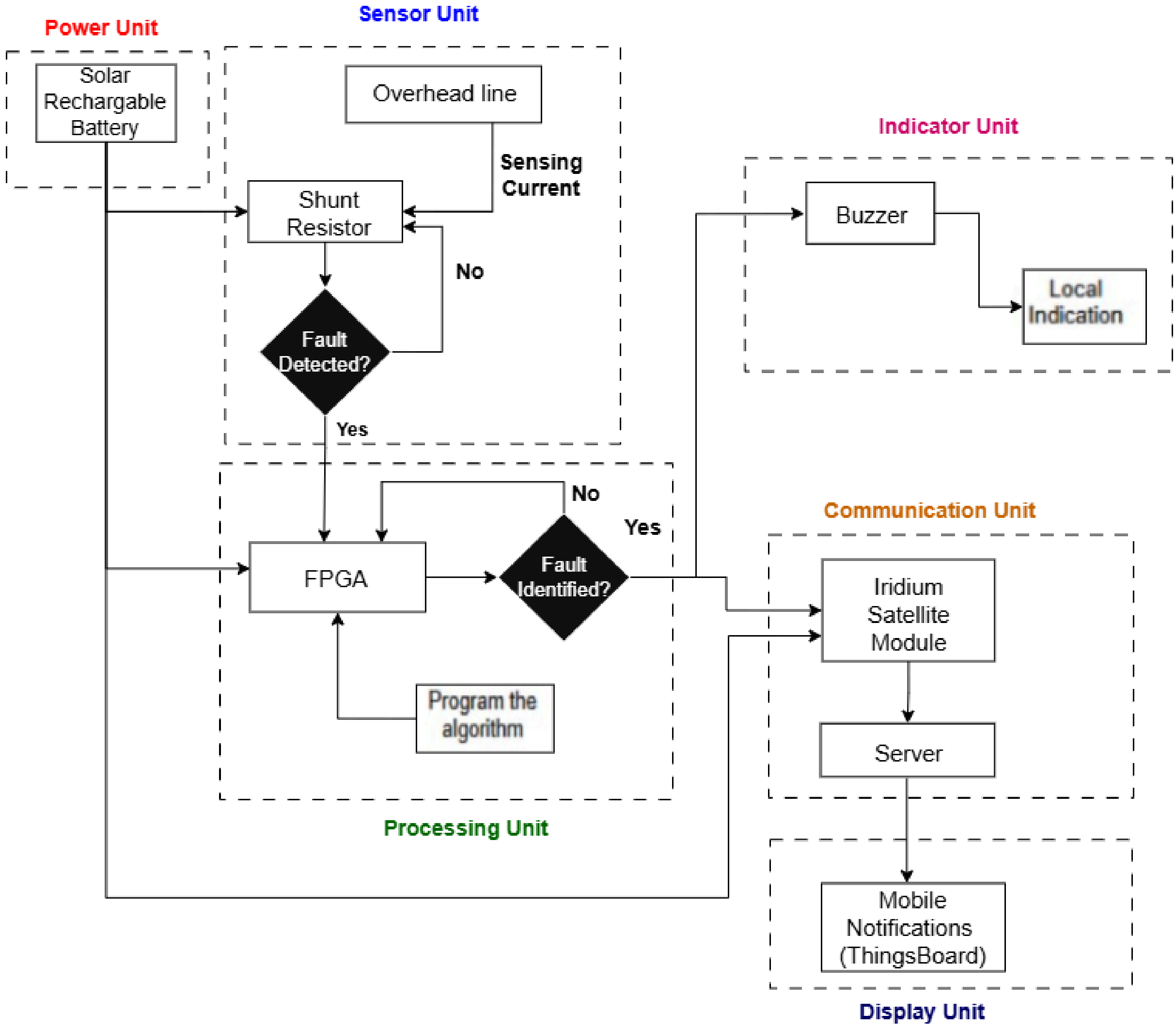


Figure 4: Block diagram for design approach 3

Design Approach 3 is a high performance Industrial graded Fault Passage Indicator, designed for use by the utility. Current sensing is achieved using Shunt Resistor, also known as current transformer (CT), which provides electrically isolated and highly accurate current measurement which are suitable to use for high current distribution lines. The conditioned signal is processed by an FPGA based processing unit which provides the high speed parallel fault detection and accurate time analysis. For communication purposes, an LTE or satellite communication module is used to ensure the data is transmitted without any issue during communication and does not rely on local network infrastructure. This allows integration with utility SCADA or centralized monitoring systems. The system is powered by a high-capacity solar panel and LiFePO4 battery system which ensure that the system can run continuously under severe environmental conditions. While this approach has the advantages of maximum reliability and performance, its high cost and complexity are drawbacks that prevent their mass deployment, so they are considered some sort of benchmark or reference design.

CHAPTER 4: PLANNING: PROJECT PLAN, BUDGET, AND OUTCOMES

4.1) Project Plan (CO11)

The expected timeline for the project is one year, which will be done over a period of three semesters. The tentative task timeline can be expressed through the Gantt chart below:

Tasks	Weeks											
	1	2	3	4	5	6	7	8	9	10	11	12
Topic Selection												
Literature Review												
Consult with Stakeholders												
Designing Multiple Approaches												
Draft Note Preparation												
Setting Tentative Budget												
Risk Assessment												
Ethics and Sustainability												
Project Proposal Report and Presentation												

Table 2(a): Gantt Chart for FYDP-P

Tasks	Weeks													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Finalise Design														
Design 1 Simulation														
Design 2 Simulation														
Design 3 Simulation														
Analysis and Compare														
Selecting the Best Approach														
Final Report and Presentation														

Table 2(b): Gantt Chart for FYDP-D

Tasks	Weeks													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Resource Sourcing														
Construction of Subsystem														
Testing and Debugging														
Subsystem Integration														
Testing and Refining														
Project Validation														
Final Project Demonstration														

Table 2(c): Gantt Chart for FYDP-C

The tasks of the project have been distributed evenly among the team. This ensures better efficiency and smooth collaboration among team members. These tasks have been assigned to the team members based on individual strengths so that everyone can work seamlessly to achieve the expected outcome. The responsibilities are outlined below:

Tasks	Responsibilities			
	Lead Contributor	Co-Contributor	Support	Reviewer
Literature Review	All	All	All	All
Report Writing	All	All	All	All
Market Research	Daiyan	Aban	Shahriar, Nidhi	All
Budget Preparation	Daiyan	Aban,Shahriar	Nidhi	All
Resources Procurement	Shahriar	Daiyan	Aban	Nidhi
Simulation of Designs	Shahriar	Aban, Daiyan	Nidhi	All
Machine Learning Algorithm Development	Daiyan	Aban	Shahriar	All
Hardware Construction	All	All	All	All
Testing and Analysis	Shahriar	Aban	Daiyan,Nidhi	All
Risk Assessment	Nidhi	Shahriar	Aban	Daiyan

Table 3: Responsibility Matrix

4.2) Budget (CO11) (Tentative)

Design Approach 1:

Item	Purpose	Unit Price (BDT)	Quantity	Total Price (BDT)
Rogoswki Coil	Current measurement for LV lines	1,800	2	3600
STM32 Microcontroller	Device control & fault logic	1,200	3	3600
GSM Module (SIM800L)	Communication with server	1,200	2	2400
Li-ion Battery (3.7V, 2200mAh)	Power supply	450	3	1350
TP4056 Charger	Battery charging	80	2	160
Voltage Regulator (LM2596)	Stable system power	150	3	450
LED + Buzzer Indicator	Local alert system	120	3	360
Enclosure (IP55 Plastic)	Outdoor protection	600	2	1200
PCB + Wiring + Connectors	Assembly materials	500	3	1500
			Total Price (Approach 1)	**14620 BDT**

Table 4: Tentative Budget for Design Approach 1

Design Approach 2

Item	Purpose	Unit Price (BDT)	Quantity	Total Price (BDT)
High-Accuracy Hall Sensor	±100A measurement	3,200	2	6400
ESP32 / Raspberry Pi Pico W	On-board analytics & control	1,500	3	4500
LoRaWAN Module (RFM95W)	Long-range communication	1,900	2	3800
Li-ion Battery (3.7V, 4000mAh)	Reliable power	1,300	2	2600
Solar Panel (5W)	Optional charging	1,400	1	1400
MPPT Charge Controller	Solar/battery management	1,150	2	2300
Weatherproof Enclosure (IP65)	Outdoor durability	1,000	2	2000
SD Card 16GB	Local data logging	900	2	1800
PCB, Connectors, Fasteners	Assembly materials	1900	3	5700
LoRa Antenna mounting kit	Better range & installation	500	2	1000
			Total Price (Approach 2)	**31500 BDT**

Table 5: Tentative Budget for Design Approach 2

Design Approach 3

Item	Purpose	Unit Price (BDT)	Quantity	Total Price (BDT)
Precision Current Transformer	High-accuracy LV sensing	3,000	2	6000
Xilinx FPGA Board (Zynq)	High-speed computation	45,000	1	45000
Satellite / LTE Industrial Modem	Reliable remote communication	22,000	1	22000
$LiFePO_4$ Battery (12V, 10Ah)	Long backup	4,500	2	9000
Solar Panel (20W)	Sustained power supply	2,500	2	5000
Solar Charge Controller	Battery protection	650	2	1300
Industrial IP67 Enclosure	Harsh environment durability	3,500	1	3500
Heat Sink + Surge Protection	Thermal & surge safety	1,200	2	2400
PCB, Connectors, Mounts	Assembly accessories	1,500	3	4500
			Total Price (Approach 2)	**98700 BDT**

Table 6: Tentative Budget for Design Approach 3

4.3) Expected Outcome

The main deliverable of the given project is that an effective design and development of a smart Fault Passage Indicator (FPI) device that consistently satisfies the demands of fault current detection and identification on power distribution networks was achieved. It is believed that the developed FPI system should independently observe the line current conditions and precisely identify the normal operating conditions and fault occurrences e.g. short-circuit, overload, and transient fault currents. The system would be characterized by a high level of detection, and a short response time hence the fault will be indicated promptly in compliance with the requirement of the normal distribution system.

Besides fault detection, the project will be able to provide a powerful communication enabled FPI that will be able to interface easily with a centralized monitoring server. It will communicate to the server using proper wired or wireless communication protocols in order to be updated about the fault status (in real-time) and the current magnitude and event timestamps. This will allow remote monitoring, improved fault localization, and shortened outage restoration, which will enhance the general reliability of the system and overall efficiency of the distribution network.

Development and Verification

The project shall lead to a complete hardware-software prototype that will consist of current sensing, signal conditioning, embedded processing and communication units. This integrated intelligence fabricated in the form of signal processing and algorithmic decision logic will enable the FPI to categorize fault conditions across a measure of functionality and concurrence. The system will be taken through rigorous laboratory tests in different simulated fault conditions, such as different magnitudes, different fault duration, and different load conditions of the system.

Measurement performance will be conducted using the results obtained in relation to set benchmarks and industry standards. The detection accuracy, response time, false alarm rate and communication reliability are the key performance indicators that are supposed to be within the acceptable threshold, which should verify the effectiveness of the proposed design.

Reliability and Practical Applicability

It is assumed that the intelligent FPI system will exhibit a consistent performance in case of repeated tests, which is the characteristics of its applicability to the real distribution environment. This modular and scalable design approach will enable future upgrades, e.g. the integration with more advanced analytics or machine-based fault classification. The identified system will allow locating the faults quickly and remotely, thus, lessening the maintenance time, enhancing the network resilience, and raising the situational awareness of the utility operators.

In general, the fulfillment of the given project will lead to the creation of an effective, communication-based smart FPI solution capable of counteracting the major issues of contemporary power distribution systems, implementing it in accordance with the smart grid programs and providing a viable background of further research and technical application.

CHAPTER 5: CREATING IMPACT AND SUSTAINABILITY

5.1) Impact (CO3)

The intelligent Fault Passage Indicator (FPI) proposed should have important technical, operation, and social impacts since it will ensure improved fault detection, safety, and reliability in the LV distribution power networks. The technology will enable rapid fault location and real-time communication and minimize human intervention in the process.

Societal Impact

From a broader societal viewpoint, it is evident that the intelligent FPI system will be able to make a considerable impact in ensuring a stable power supply in Bangladesh. This is because, despite having excess power, in both DPDC and DESCO, consumers are faced with regular power interruptions because of distribution network-related inefficiencies [21], [22].

Through faster fault detection and restoration, the proposed system will ensure a reduction in outage occurrences, hence improving the standard of living as well as economic productivity in urban, as well as rural, areas. Service reliability will also enhance confidence in utility service delivery, hence facilitating inclusive development in Bangladesh [23].

Health Impact

Availability of power is important to health institutions, especially hospitals and emergency centers which require continuous power to sustain life through various appliances. Power faults in the supply chain may compromise the welfare of patients as well as the continuity of health operations [25]. The smart FPI system prevents this by shortening the time to power restoration.

Additionally, the system improves occupational health conditions for utility workers. Manual inspection of faults has occupational health hazards such as electrocution, arc flash, and heat stroke. The use of remote indication for faults has greatly reduced these occupational health dangers, hence improving health and safety at work [23].

Safety Impact

Safety upgrades form a central impact of the proposed system. Old and overburdened distribution networks in the DPDC, DESCO, and BREB power utilities tend to contribute to the risk of power faults, wildfires, and equipment breakdowns [23]. The intelligent FPI system improves the safety of operation by allowing real-time faults detection along with proper localization of faults.

It will reduce the need for technicians to patrol energized lines when faults are present. This will result in a reduced risk of incidents due to accidents. Fast isolation of faults will also reduce secondary hazards such as conductor breakage, transformer faults, and electrical fires that normally occur in a populated region [21].

Legal Impact

Legally and regulatory-wise, the intelligent FPI solution is helpful for ensuring reliability and service quality requirements as a regulatory mandate of the Bangladesh Energy Regulatory Commission (BERC). The complaint ratio involving consumers and the issue of power outages between DPDC and DESCO establish the need for proper documentation of power outages [24].

Automated logging of faults and time-stamped data can enhance accountability, facilitate regulatory statistical reports, and minimize possibilities for litigation regarding prolonged downtime and services. The system can enhance utilities' capacity to meet legal requirements and regulatory technical performance standards [24].

Cultural Impact

Culturally, power cuts have long been accepted in Bangladeshi culture, especially in rural and semi-urban parts. But with increased urbanization and dependence on digital technology, customer demand for improved power service is expected to rise [21], [25]. The use of intelligent FPI systems contributes to bringing about a new customer culture in power services with preventive maintenance.

Moreover, the system promotes a professional environment favorable to ensuring the safety of engineers and technicians, as it does not encourage risk-taking involving manual fault detection. This approach also conforms to contemporary professional ethics among engineers and impacts the modernization of DPDC, DESCO, and BREB institutions accordingly [22], [23].

5.2) Sustainability (CO4)

In this project, sustainability is assessed as meeting long-term environmental responsibility, economic viability, and social resilience. The smart FPI system is engineered to function effectively to satisfy current requirements of the distribution network without sacrificing the ability to meet the needs of the future. One of the initial steps in assessing the sustainability of a project is SWOT analysis, which stands for Strengths, Weakness, Opportunities and Threats, and is given below for this project:

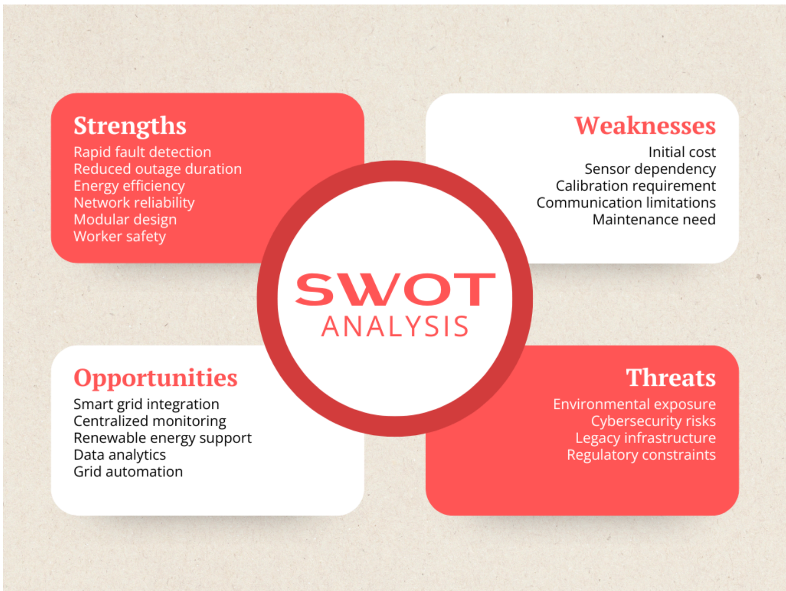


Figure 5: SWOT Analysis for intelligent FPI system

The sustainability of design approaches' methodologies can be validated in three types: Environmental, Economic and Social aspects, in order to understand how viable and realistic they are in judging overall performance in sustainable expectations. These include:

Environmental Sustainability

The proposed FPI system is environmentally sustainable since it supports efficient energy distribution and avoids unnecessary power waste in case of a fault. The low-power sensing and communication design of the system ensures that it consumes low energy in its continuous operation. The system is environmentally sustainable since it avoids environmental degradation associated with discarding distribution equipment and frequent replacement with new components.

Moreover, this system is capable of supporting interoperations with smart grid systems, which are very important with regard to controlling renewable energy resources and further development in grid infrastructure.

Relevant Sustainable Development Goals:

- SDG 7 – Affordable and Clean Energy: Achieved by improving energy efficiency, reducing losses during fault conditions, and supporting reliable power distribution.
- SDG 12 – Responsible Consumption and Production: Addressed by extending equipment lifespan and minimizing electronic waste through preventive fault detection.
- SDG 13 – Climate Action: Supported indirectly by reducing energy losses and operational emissions associated with prolonged faults and maintenance activities.

Economic Sustainability

Considering economic sustainability, it is evident that the intelligent FPI provides a good balance between performance and affordability. This is due to its use of commercial sensors, microcontrollers, and communication modules, which make it affordable. Moreover, its scalability enables utilities to employ a scale-by-budget strategy.

Lowered maintenance expenditure, reduced outage charges, and increased equipment life span are factors that contribute to its long-term economic viability. Also, it promotes skill development within its community in terms of embedded systems and power system monitoring. This assists in ensuring sustainability of the workforce within the energy industry.

Relevant Sustainable Development Goals:

- ❖ SDG 8 – Decent Work and Economic Growth: Achieved by supporting skilled technical employment and improving productivity within the power sector.
- ❖ SDG 9 – Industry, Innovation and Infrastructure: Addressed through the deployment of intelligent monitoring solutions that enhance the resilience and efficiency of power distribution infrastructure.

Social Sustainability

Social sustainability is addressed through the improvement of grid reliability, safety, and accessibility. Reliable fault detection provides equitable access to stable electricity-the foundation necessary for economic activity, education, healthcare, and digital connectivity. The system instills public trust in utilities and infrastructure through a reduction in outage frequency and duration.

Another important intelligent FPI feature is the reduction of human exposure to dangerous fault conditions through remote monitoring and diagnostics, promoting safer working conditions for utility personnel and, by that virtue, ethical engineering practices. In the long run, the deployment of such an intelligent monitoring system will provide a resilient and inclusive energy infrastructure that can absorb increasing demand and evolving societal needs.

Relevant Sustainable Development Goals:

- ❖ SDG 3 – Good Health and Well-Being: Achieved by reducing occupational hazards and minimizing exposure to dangerous electrical fault conditions.
- ❖ SDG 10 – Reduced Inequalities: Supported by ensuring reliable electricity access across different regions and communities.
- ❖ SDG 11 – Sustainable Cities and Communities: Addressed by strengthening critical urban infrastructure and improving the reliability of essential services such as electricity.

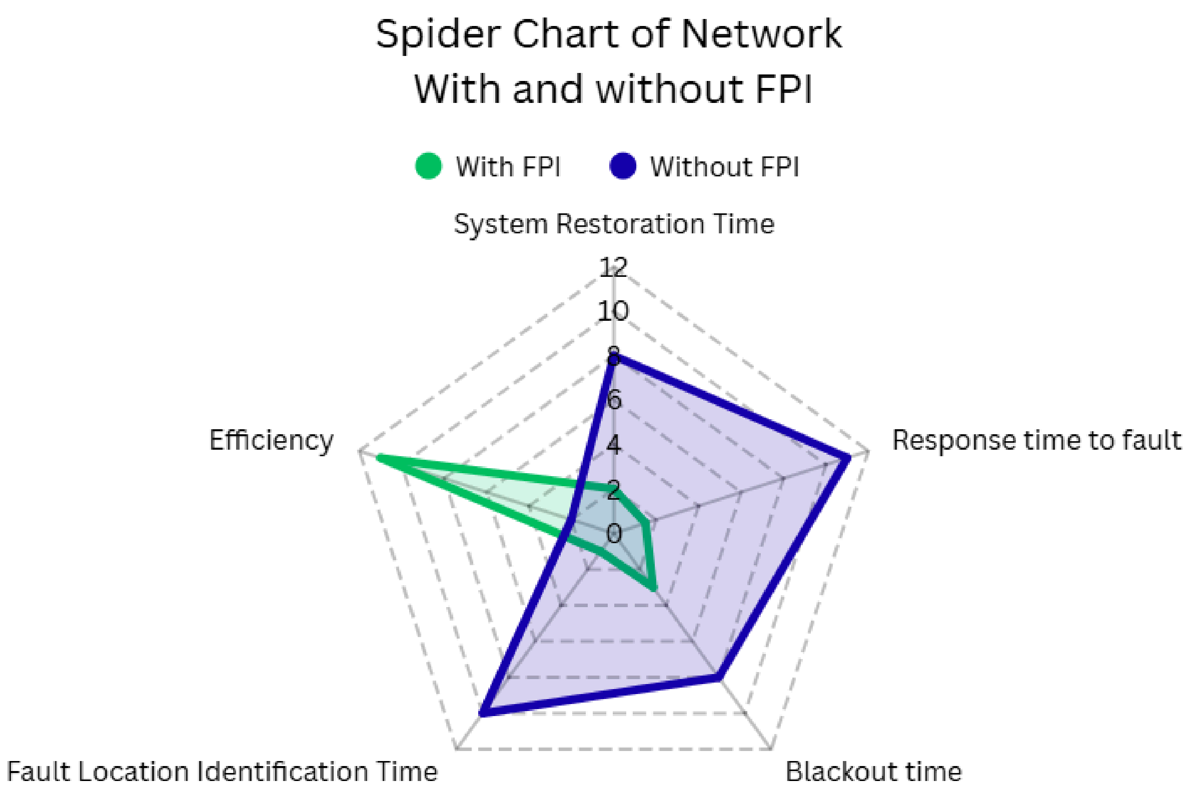


Figure 6: Spider Chart of Network with and without FPI

The Above Spider chart shows that a system with FPI is sustainable and it ensures reduction in various critical parameters like blackout time , location identification and response time and increases efficiency.

CHAPTER 6: UPHOLDING STANDARDS THROUGH CODES AND ETHICS

6.1) Applicable Standards and Codes (CO2)

Category	Standard / Code	Description	Relevance to FPI
Power System Protection	**IEEE C37.113**	Guide for Protective Relay Applications to Distribution Lines	Widely used reference in Bangladesh for distribution protection logic and fault detection principles
	IEC 60255	Measuring Relays and Protection Equipment	Applicable to fault detection behavior and protection device performance
Low-Voltage Distribution & Safety	**Bangladesh Electricity Rules (BER), 1937 (Amended)**	National Electrical Safety and Operation Rules	Primary legal framework governing LV distribution safety in Bangladesh
	IEC 60364	Low-Voltage Electrical Installations	Commonly followed international standard for LV installations and safety
	IEEE 141 (Red Book)	Electric Power Distribution Systems	Used in academic and utility studies for LV distribution behavior
Current Measurement & Sensors	**IEC 61869-2**	Instrument Transformers – Current Transformers	Relevant for accuracy and insulation requirements of current sensing
	IEEE C57.13	Instrument Transformer Standards	Provides accuracy and safety guidelines for current measurement
	IEC 61010-1	Safety Requirements for Electrical Measurement Equipment	Ensures operator and equipment safety
Communication & IoT	**IEEE 802.11**	Wireless LAN (Wi-Fi) Standard	Commonly used for local communication and testing
	LoRaWAN Specification	Low-Power Wide Area Network	Suitable for rural and semi-urban Bangladesh
	3GPP GSM / LTE	Cellular Comms Standards	Nationwide and practical for utility monitoring

Category	Standard / Code	Description	Relevance to FP
Electromagnetic Compatibility (EMC)	**IEC 61000-6-2**	EMC Immunity for Industrial Environments	Ensures immunity against EMI in substations and distribution areas
	IEC 61000-6-4	EMC Emission Standards	Limits electromagnetic interference to nearby equipment
	IEEE 519	Harmonic Control in Power Systems	Relevant due to non-linear loads and power quality issues in Bangladesh
Environmental & Mechanical Protection	**IEC 60529**	Degrees of Protection (IP Code)	Defines enclosure protection (IP55–IP65) for outdoor installation
	IEC 60068	Environmental Testing	Ensures operation under heat, humidity, and dust conditions
Cybersecurity & Data Integrity	**IEEE 1686**	Cyber Security for Intelligent Electronic Devices	Increasingly relevant for smart grid and IoT-based monitoring.
Ethics & Sustainability	**IEEE Code of Ethics**	Professional Engineering Ethics	Used in engineering education and professional practice.
	UN SDGs (7, 9, 11)	Sustainable Development Goals	Aligns with Bangladesh's national energy and infrastructure goals

Table 7: Table for Applicable Standards and Codes

6.2) Ethical Consideration (CO13)

The Intelligent Fault Passage Indicator (FPI) project addresses several ethical considerations to ensure responsible engineering practice, public safety, and long-term societal benefit:

1. Reliability and Accuracy of Fault Detection

A primary ethical concern is the reliability and accuracy of the fault indication process, as incorrect fault indications or failures to detect faults can lead to prolonged outages, equipment damage, or safety hazards for personnel and the public. To mitigate these risks, the system is designed with rigorous testing, validation, and benchmarking against established performance criteria. Compliance with standards such as IEEE C37.113 – IEEE Guide for Protective Relay Applications to Distribution Lines ensures that fault detection logic and sensitivity meet industry-accepted performance benchmarks. Additionally, IEEE Std 519 guidelines on system harmonics and signal quality support accurate measurement interpretations under a range of power system disturbances. Regular validation against these standards promotes dependable operation across diverse fault conditions.

2. Data Integrity, Confidentiality, and Secure Communication
Data integrity and confidentiality are critical ethical aspects of the FPI, particularly given the communication between the FPI device and the central monitoring server. Fault data and system status information must be transmitted securely to prevent unauthorized access, tampering, or misinformation that could compromise system operations or safety. Ethical responsibility dictates the implementation of secure communication protocols and access controls conforming to security standards such as IEEE Std 1686 – IEEE Standard for Intelligent Electronic Devices (IEDs) Cyber Security Capabilities, and IEEE 802.1X – Port-Based Network Access Control for authentication in networked environments. These standards establish baseline requirements for secure device operation, encrypted data exchange, and controlled access to system information, ensuring that only authorized personnel can access sensitive data.

3. Workforce Safety and Responsibility
The FPI's ability to enable remote fault detection and monitoring directly supports ethical obligations toward workforce safety by minimizing the need for manual inspections in hazardous environments. By reducing personnel exposure to live electrical equipment, elevated structures, and other physical risks, the system aligns with general safety engineering principles and complements standards such as IEEE 1584 – Guide for Performing Arc-Flash Hazard Calculations which informs safe work practices on energized systems. Maintaining a design focus on minimizing human exposure to risk reflects a commitment to protecting field engineers and maintenance staff from unnecessary danger.

4. Ethical Engineering Practice, Standards Compliance, and Documentation
Ethical engineering practices are upheld through diligent compliance with applicable standards, responsible component selection, and transparent documentation of design decisions and system limitations. Following standards such as IEEE 1547 – Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces ensures that the system behaves predictably within broader grid operations and interoperability frameworks. Transparent documentation of design assumptions, test results, failure modes, and maintenance procedures supports accountability and enables future engineers to understand and extend the system safely. Emphasis on sustainability, reliability, and accountability ensures that the developed system serves the public interest without compromising safety or professional integrity.

CHAPTER 7: RISK ASSESSMENT AND ANALYSIS

7.1) Risk Management and Analysis

Effective risk management is necessitated in ensuring the successful development and implementation of the intelligent FPI system. Risks that could result from technical performance, operability, and environmental factors have also been identified and assessed. The fault tree analysis of the project has been illustrated in Figure 7.

Fault Tree Analysis:

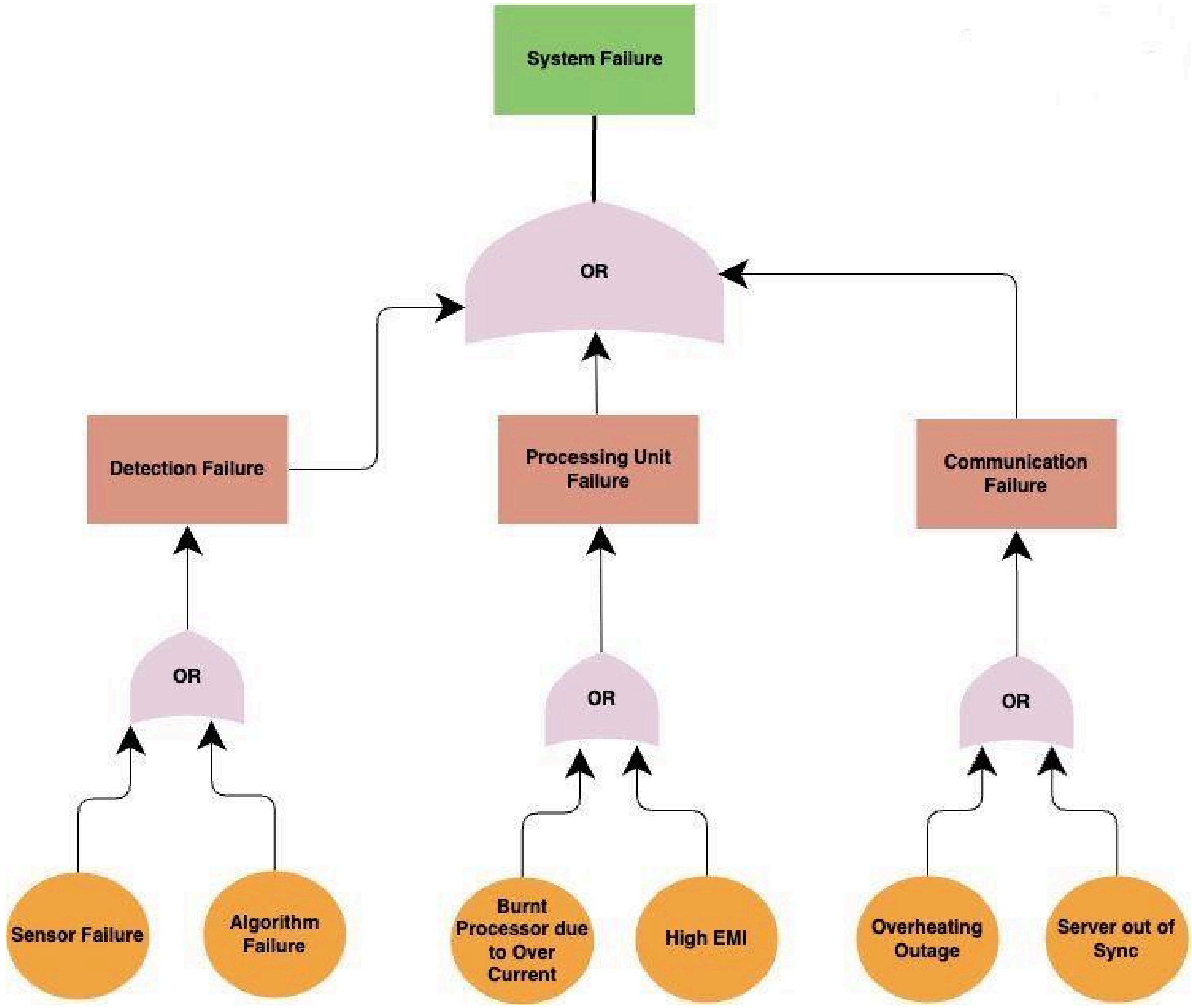


Figure 7: Fault Tree Analysis

Risk Analysis:

Risk analysis will identify all possible technical, operational, environmental, and managerial risks that could affect the successful design, development, and validation of the intelligent FPI system for low-voltage distribution networks. Because the project covers hardware sensing, embedded processing, communication modules, and power electronics, several uncertainties can be expected at different stages of the development lifecycle. The following are some key risks that could impact the project:

1. Inaccurate fault detection due to load fluctuations (R1): Low-voltage distribution networks are subjected to frequent current variations due to changing consumer loads. These fluctuations may appear as fault signatures and hence enhance the likelihood of false fault detection or missed fault events. Such inaccuracies could diminish the effectiveness and credibility of the FPI system.

2. Current sensor malfunction or measurement inaccuracy (R2): The current sensors that use the Hall effect principle can be affected by noise, electromagnetic interference, drift effects due to temperature changes, as well as poor location. This means that any error in the current sensors can directly cause inaccurate current readings.

3. Communication network failure or data latency (R3): The system uses the wireless communication technologies of LoRaWAN, GSM, or Wi-Fi for fault data transmission. For example, network unavailability, packet loss, or high latency-in particular, in villages or other rural areas-reduces fault reporting and will lower the benefit of real-time monitoring.

4. Power supply and battery-related failure (R4): In some cases, since the FPI is battery-powered and has the option of being powered using a solar charger, a lack of charging or improper battery or power management functions could cause the system to shut down.

5. Hardware integration and compatibility issues (R5): The FPI integrates multiple hardware components, including sensors, microcontrollers, communication modules, and power units. In the case of incompatible components, errors in wiring, or interface mismatches, the system can also run into instability or complete failure during operations.

6. Environmental and weather-related damage (R6): Outdoor installation means that the system will be subjected to high temperature, humidity, rain, dust, and environmental pollution. All these environmental conditions can cause malfunctioning and reduce the life span of devices and components.

7. Safety hazards during installation and testing (R7): When working in the area of live low-voltage power lines, there could be possible risks of electric shock, short circuit, or damage. Inappropriate insulation, earthing, and handling practices while conducting the test could cause potential dangers to the operator as well as the equipment.

8. Component availability issues (R8): Some electronic components, for example, current sensors, communication modules, and microcontrollers, may be susceptible to supply chain disruptions or local market shortages. It may become necessary to substitute these components, and change the design.

9. Budget overrun (R9): Unexpected price increases for project components, additional test requirements, or the replacement of defective parts may result in the total cost of the project going over the budget. It may also result in cutting down the functionality of the system or using lower-quality components.

10. Regulatory approval delays (R10): Use of electrical monitoring devices in the electrical power distribution systems may call for the approval of the concerned authority or the corresponding electrical standards in the country. Lack of approval or non-compliance with the electrical standards may cause delays.

Fever Chart :

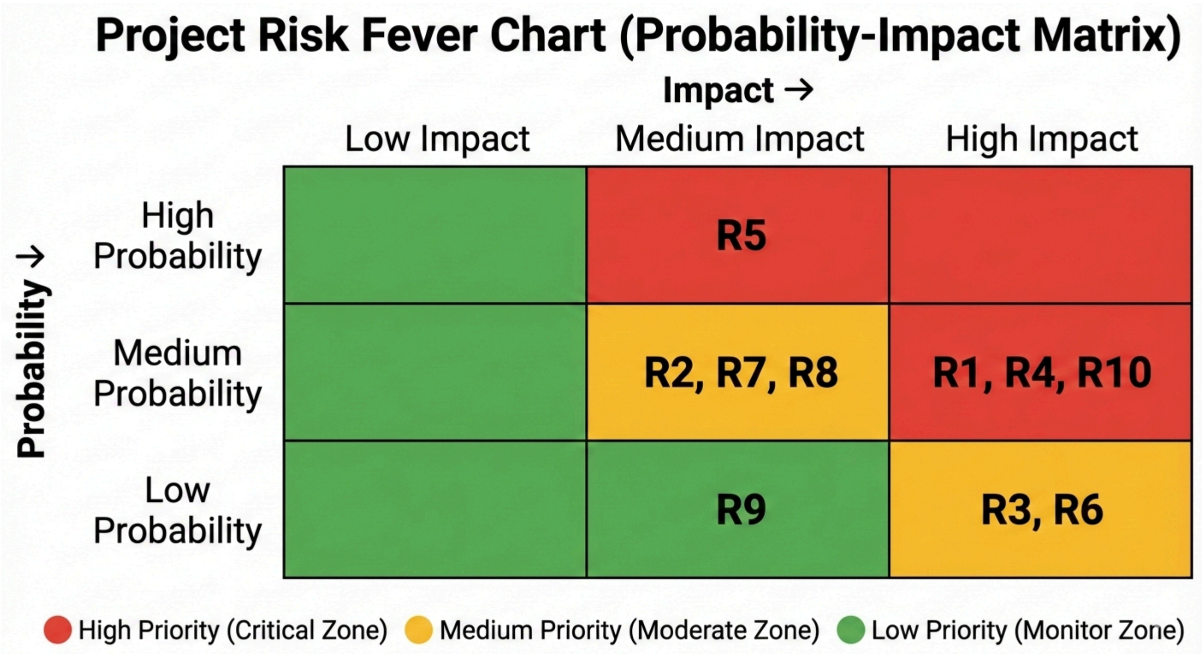


Figure 8: Fever Chart for Qualitative Assessment of FPI

Risk Management Plan:

The risk management plan outlines the preventive and corrective measures to reduce the effect of technical, operation, environment, and managerial risks that are associated with the intelligent Fault Passage Indicator (FPI) system.

Inaccurate Fault Detection Due to Load Fluctuations(R1):

- **Preventive Measures:** Effective fault detection algorithms would be developed using a set of parameters including RMS current value, rate of change, and duration of the fault. These thresholds would be chosen upon a literature study, along with low-voltage faults.
- **Mitigation Strategy:** Methods of digital filtering and adaptive thresholding will be used to identify whether it is a normal load condition or a possible faulty condition. Laboratory testing with different load profiles is to be performed to improve the results.
- **Contingency Plan:** If there is continued false detection, there will be improvements to detection logic and sensitivity levels, and possibly the addition of further faulty indicators such as time validation.

Current Sensor Malfunction or Inaccuracy (R2):

- **Preventive Measures:** High-quality Hall-effect sensors that meet current and temperature specifications will be used. The sensor installation instructions and adjustments must be followed before it is used.
- **Mitigation Strategy:** Noise reduction algorithms in software will be employed along with periodical calibration procedures for reducing sensor drift and electromagnetic interference.
- **Contingency Plan:** In the case of a sensor malfunction, substitute sensors or another method of sensing would be utilized to continue the testing and validation process of the system.

Communication Network Failure or Latency (R3):

- **Preventive Measures:** Various communications technologies such as LoRaWAN, GSM, or Wi-Fi will be supported for flexibility.
- **Mitigation Strategy:** Buffering and re-transmission methods for data will be incorporated to ensure that no data is lost when there is a network downtime or latency.
- **Contingency Plan:** In case the communication failure continues, the events of faults will be recorded locally and retrieved during the maintenance or test phases.

Power Supply and Battery Failure (R4):

- **Preventive Measures:** Low power components will also be selected, and power-efficient firmware design will be used to reduce power consumption.
- **Mitigation Strategy:** Battery voltage measurement and sleep mode during idle times will be enabled to improve the lifetime of the device.
- **Contingency Plan:** In the case of the battery running out, battery recharging, as well as other external sources of power, will be utilized for the test process.

Hardware Integration Issues (R5):

- **Preventive Measures:** A modular system design will be adopted, which will enable testing of a subsystem like sensing, processing, communication, and power independently.
- **Mitigation Strategy:** The process of incremental integration and debugging will be followed to identify and fix interface incompatibilities or wire problems.
- **Contingency Plan:** In the event that problems with integration occur, simplified subsystem arrangements will be employed for verification of basic functionality.

Environmental and Weather-Related Damage (R6):

- **Preventive Measures:** Weather-tight enclosures such as at least IP55 can be used for protecting components against moisture, dust, and temperature.
- **Mitigation Strategy:** Protective insulation, grounding, and surge protection devices will also be introduced in order to withstand environmental stress.
- **Contingency Plan:** In the situation of environmental degradation, the affected components would be replaced and the testing would take place in a controlled laboratory environment.

Safety Hazards During Installation and Testing (R7):

- **Preventive Measures:** All the safety precautions for the low voltage electrical system are followed including insulation, grounding, and personal protection equipment.
- **Mitigation Strategy:** Initial testing under simulated laboratory conditions will be conducted before handling live distribution lines.
- **Contingency Plan:** If there are any safety risks, testing will be stopped immediately and reviewed under supervision before resumption.

Component Availability (R8):

- **Preventive Measures:** The components will be selected based on market availability in the locality. Equivalent alternative components will be located during the design phase to ensure that there is not just one source for obtaining the component.

- **Mitigation Strategy:** Early procurement planning and having a list of substitute components will help reduce delays. Flexibility of design will be incorporated to permit substitution of components that do not impair system functionality.
- **Contingency Plan:** In case certain parts turn out to be unavailable, equivalent substitute parts with similar parameters and features would be used.

Budget Overrun (R9):

- **Preventive Measures:** The initial stage will involve the preparation of a very detailed and realistic budget, with cost buffers included for unexpected expenses.
- **Mitigation Strategy:** It will also involve regular budget tracking, monitoring of the costs involved throughout the project life cycle, using cost-effective components, and open-source platforms wherever possible.
- **Contingency Plan:** When there's an overrun in the budget, non-critical features will be postponed or excluded, and the system will be optimized to demonstrate core functionality within the available budget.

Regulatory Approval Delays (R10):

- **Preventive Measures:** Significant national standards, safety regulations, and requirements for low voltage distribution systems in Bangladesh would be considered at the beginning of the design process.
- **Mitigation Strategy:** The system designs shall be made in accordance with existing IEEE, IEC, and Bangladesh Electricity Rules to reduce any issues regarding system compliance.
- **Contingency Plan:** In case of a delay in approval, it will move forward to a laboratory validation process as well as a simulation study pending field approval.

CHAPTER 8: ENSURING SAFETY ASPECTS

8.1) Safety Consideration (CO3)

Safety is an equally important consideration in the intelligent FPI system since it is intended to be used in the electrical distribution sector. Electrical safety is ensured by proper insulation, grounding, and isolation of the sensing or processing circuit from the high-voltage power lines. Current sensing components are chosen that ensure proper measurement of the current signals with minimal risk of electrical shock.

It improves operational safety as fewer persons will be required to go out and inspect for faults manually. Remote monitoring with fault indication greatly limits the human exposure to high-risk zones during any fault conditions. The system allows fail-safe operation in order to ensure that, in case of an internal malfunction, it does not introduce any additional hazard either to the power network or personnel.

Environmental safety considerations include weatherproofing, electromagnetic compatibility, and thermal management to prevent overheating or unintended emissions. The system complies with relevant safety and electromagnetic exposure standards to ensure safe operation around humans and animals.

Overall, the design integrates preventive and protective safety measures to ensure that the FPI system operates reliably without compromising human safety or network integrity.

CHAPTER 9: CONCLUSION

This project presents the design and development of an intelligent Fault Passage Indicator (FPI) system aimed at enhancing fault detection, monitoring, and reliability in power distribution networks. By integrating current sensing, embedded processing, and communication capabilities, the proposed system enables accurate fault identification and real-time reporting to a central monitoring server.

The project emphasizes reliability, safety, and sustainability through rigorous testing, ethical engineering practices, and risk-aware system design. Laboratory validation across multiple fault scenarios ensures that the system meets defined performance benchmarks and operational requirements. The communication-enabled architecture supports faster fault localization, reduced outage duration, and improved maintenance efficiency.

In conclusion, the intelligent FPI system offers a practical and scalable solution aligned with modern smart grid initiatives. Its successful implementation can significantly improve distribution network resilience, operational safety, and service reliability, while providing a strong foundation for future enhancements such as advanced analytics and intelligent grid automation.

CHAPTER 10: APPENDIX AND REFERENCES

10.1) Attributes of Complex Engineering Problems (EP)

	Attributes	**Put tick (√) as appropriate**
P1	Depth of knowledge required	√
P2	Range of conflicting requirements	√
P3	Depth of analysis required	√
P4	Familiarity of issues	
P5	Extent of applicable codes	√
P6	Extent of stakeholder involvement and needs	√
P7	Interdependence	√

Note: Project must have P1, and some or all from P2-P7

10.2) Attributes of Complex Engineering Activities (EA)

	Attributes	**Put tick (√) as appropriate**
A1	Range of resource	√
A2	Level of interaction	√
A3	Innovation	
A4	Consequences for society and the environment	√
A5	Familiarity	

Note: Project must have some or all of the characteristics from attributes A1 to A5

10.3) Glossary of Terms

ACSR-Aluminum Conductor Steel Reinforced.

ADMS-Advanced Distribution Management System.

ATC-Academic Technical Committee.

BERC-Bangladesh Energy Regulatory Commission.

BPDB-Bangladesh Power Development Board.

CAPEX-Capital Expenditure.

CO-Course Outcome.

CT-Current Transformer.

DESCO-Dhaka Electric Supply Company.

di/dt-Rate of Change of Current.

DPDC-Dhaka Power Distribution Company.

ENS-Energy Not Supplied.

FPI-Fault Passage Indicator.

FLISR-Fault Location Isolation and Service Restoration.

FPGA-Field-Programmable Gate Array.

FYDP-Final Year Design Project.

GSM-Global System for Mobile.

HECS-Hall-effect Current Sensor.

HIEF-High-Impedance Earth Fault.

IEC 60255-Measuring Relays/Protection Equipment.

IEC 60364-Low-Voltage Electrical Installations.

IEC 61000 Series-Electromagnetic Compatibility.

IEEE C37.113-Protective Relay Guide for Distribution Lines.

IEEE C57.13-Instrument Transformer Standards.

IEEE Std 1686-IED Cyber Security Capabilities.

IEEE Std 519-Harmonic Control in Power Systems.

IP55/IP65/IP67-Ingress Protection Rating.

LiFePO4-Lithium Iron Phosphate Battery.

LoRaWAN-Long Range Wide Area Network.

LTE-M-LTE for Machines.

LV-Low Voltage.

MTBF-Mean Time Between Failures.

MTTD-Mean Time to Detection.

MV-Medium Voltage.

OMS-Outage Management System.

OPEX-Operational Expenditure.

PD-Partial Discharge.

PO-Program Outcome.

PSDP-2041-Power Sector Development Plan to 2041.

RDSS-Revamped Distribution Sector Scheme.

SBC-Single-Board Computer.

SAIDI-System Average Interruption Duration Index.

SCADA-Supervisory Control and Data Acquisition.

SDG-Sustainable Development Goal.

SLG-Single-Line-to-Ground.

STM32-ARM Cortex-M Microcontroller.

SWOT-Strengths Weaknesses Opportunities Threats.

THD-Total Harmonic Distortion.

VT-Voltage Transformer.
WZPDCL-West Zone Power Distribution Company Limited.

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10.5) LogBook of FYDP-P

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Date/Time /Place	Attendee	Summary of Meeting Minutes	Responsible	Comment by ATC
15.03.2025	1.Aban 2.Daiyan 3.Shahriar 4.Nidhi 1.Belal Bhuian Sir	Introductory Meeting 1.Finding a suitable FYDP Topic.	All four members.	N/A as it was an introductory meeting.
19.10.2025	1.Aban 2.Shahriar 3.Nidhi 1.Belal Bhuian Sir	1.Transmission Line Inspection Drone idea approached 2.Need to find Stakeholders 3.Do background study 4.Find Scope	All three members	1. To talk to people in DPDC/PDB about the kind of problems they face, the type of solutions we propose to provide and do they really need it. 2. To talk to Shameem Ahmed Sir about the

				feasibility of our plans and project.
21.10.2025	1.Aban 2.Daiyan 1. Sabbir Ahmed Sir	Discussed the updates on our topics.	Aban, Daiyan	
26.10.2025	1.Aban 2.Nidhi 3.Daiyan 1.Belal Bhuian Sir	1.The idea of Fault Passage Indicator (FPI) proposed to sir. 2.Doing the Functional Analysis 3.Specifications 4.Narrowing down the parameters	Aban, Nidhi, Daiyan	1.Follow the rubrics to write down the scopes, specifications and other things that are mentioned there.
2.11.2025	1.Aban 2.Shahriar 3.Nidhi 1.Belal Bhuian Sir 2.Mahmudul Islam Sir	1.Preparing the multiple design approach matrix 2.Finding relevant papers for all the subsystems	Aban, Shahriar, Nidhi	1.Finding relevant and correct conference/ journal papers for the entire project and specific subsystems 2.Emphasising on finding journal papers 3.Finding papers relevant to the subsystem design 4.Finding Review Papers on FPI
4.11.2025	1.Aban 1.Sabbir Ahmed Sir	1.Redesigning the serial communication of FPIs 2.Finding and giving references of specifications and numeric values of our design	Aban	1.Serial communication will not work in case of a single FPI failure. Each FPI sending the data directly would be more robust. 2.Serial communication would work if each FPI data is used by the other. 3.Working on the specification.
9.11.2025	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian sir	2. Discussion on flowchart, block diagrams and specifications.	All four members	1. Writing according to Action verb format. 2. Developing the flowchart and giving

	2. Mahmudul sir			section-wise block diagrams. 3. Ensuring different specifications for system and subsystem.
11.11.2025	1. Shahriar 1. Belal Bhuian sir	1. Updates on DPDC visit.	Shahriar	1. Gave update about visit to DPDC, necessary documents required for the project etc.
18.11.2025	1. Daiyan 1. Sabbir Ahmed sir	1. Analysis of substation data 2. Design approaches against each of the subsystems.	Daiyan	1. Talked about how to analyse substation's data using statistical methods/ML-based approaches. 2. Checked the design approaches against each of the subsystems.
5.12.2025	1. Daiyan 2. Shahriar 3. Nidhi 1. Mahmudul sir	1. Specifications 2. Functional and non-functional requirements 3. Design Approaches.	1. Shahriar 2. Nidhi 3. Daiyan	1. Specifications should be made based on system performances. 2. Writing the functional and non-functional requirements according to the specific objectives. Also including quantitative data. 3. Design Approaches.
7.12.2025	1. Shahriar 2. Aban 1. Belal Bhuian sir 2. Mahmudul Sir	1. Discussion on Design Approaches and Scopes	Shahriar Aban	1. Approaches will be based on different components and their working principles. 2. Writing Scope as a different chapter.
14.12.2025	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Belal Bhuian sir	Progress Presentation	All four members	1. Concise the title 2. Background research to be about fault, subsystem, theft

				issues and stakeholders. 3. Problem statement according to class lecture. 4. Functional and Non-Functional Requirements 5. Doing the functional analysis and then design approaches according to class lecture.
28.12.2025	1. Aban 2. Daiyan 3. Shahriar 4. Nidhi 1. Mahmudul sir	Demo presentation before the final one.	All four members	1. Editing the slides containing problem statements, functional and non-functional requirements, design approaches, gantt chart, budget etc.
30.12.2025	1. Aban 2. Shahriar 1. Belal Bhuian sir	Showed the final presentation slides.	Aban Shahriar	1. Edit the problem statement.

Assessment Guideline for Faculty

[The following assessment guideline is for faculty ONLY. **This portion is not applicable for students**.]

Assessment Tools and CO Assessment Guideline

	Distribution of assessment points among various COs assessed in different semesters														
PO □	l	c	f	g	c	b	d	c	e	l	k	k	h	i	j
CO □	CO 1	C O 2	CO 3	CO 4	CO 5	CO 6	CO 7	CO 8	CO 9	CO 10	CO 11	CO 12	CO 13	CO 14	CO 15
EEE 400P/ ECE 402P (out of 100)	15	21	15	15							15		6	6	7
Project Concept Note	x	x													x
Project Proposal Report	x	x	x	x							x				x
Progress Presentation											x				x
Peer-evaluatio n*														x	
Instructor's Assessment*											x		x	x	

Note: The star (*) marked deliverables/skills will be evaluated at various stages of the project.

Mapping of CO-PO-Taxonomy Domain & Level- Delivery-Assessment Tool

Sl.	CO Description	PO	Bloom's Taxonomy Domain/Level	Assessment Tools
CO1	**Identify** a solvable complex engineering problem preferably relevant to the current and future industry through appropriate research	l	Cognitive/ Understand	● Project Concept Note
CO2	**Identify** the objectives, specifications, functional and non-functional requirements, and constraints as well as applicable compliance, standards and codes of practice to the solution of the engineering problem	c	Cognitive/ Apply	● Project Concept Note
CO3	**Assess** the impact of the solution of the engineering project in terms of societal, health, safety, legal and cultural context	f	Cognitive/ Evaluate	● Project Proposal Report
CO4	**Evaluate** the sustainability and impact of solution of the proposed project in terms of environmental consideration	g	Cognitive/ Evaluate	● Project Proposal Report
CO11**	**Demonstrate** project management skill in various stages of developing the solution of engineering design project	k	Cognitive/ Apply Affective/ Valuing	● Project Proposal Report ● Project Progress presentation
CO13	**Apply** ethical considerations and professional responsibilities in designing the solution and throughout the project development phases	h	Cognitive/ Apply Affective/ Valuing	● Peer-evaluation , ● Instructor's Assessment
CO14**	**Perform** effectively as an individual and as a team member for successfully completion of the project	i	Affective/ Characterization	● Peer-evaluation ● Instructor's Assessment
CO15**	**Communicate** effectively through writings, journals, technical reports, deliverables, presentations and verbal communication as appropriate at various stages of project development	j	Cognitive/ Understand Psychomotor/ Precision Affective/ Valuing	● Project Concept Notes, ● Project Proposal Report ● Progress Presentations,

Note: The double star (**) marked CO will be assessed at various stages of the project through indirect deliverables.